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What Representational Similarity Measures Imply about Decodable Information

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Abstract

Neural responses encode information that is useful for a variety of downstream tasks. A common approach to understand these systems is to build regression models or "decoders" that reconstruct features of the stimulus from neural responses. Popular neural network similarity measures like centered kernel alignment (CKA), canonical correlation analysis (CCA), and Procrustes shape distance, do not explicitly leverage this perspective and instead highlight geometric invariances to orthogonal or affine transformations when comparing representations. Here, we show that many of these measures can, in fact, be equivalently motivated from a decoding perspective. Specifically, measures like CKA and CCA quantify the average alignment between optimal linear readouts across a distribution of decoding tasks. We also show that the Procrustes shape distance upper bounds the distance between optimal linear readouts and that the converse holds for representations with low participation ratio. Overall, our work demonstrates a tight link between the geometry of neural representations and the ability to linearly decode information. This perspective suggests new ways of measuring similarity between neural systems and also provides novel, unifying interpretations of existing measures.

1 Introduction

The computational neuroscience and machine learning communities have developed a multitude of methods to quantify similarity in population-level activity patterns across neural systems. Indeed, a recent review by Klabunde et al. [19] catalogues over thirty approaches to quantifying representational similarity. Many of these measures quantify similarity in the *shape* or *representational geometry* of point clouds. For example, recent papers (e.g. [35, 9]) leverage the Procrustes distance and other concepts from *shape theory*—an established body of work that formalizes the notion of shape and ways to measure distance between shapes [18, 10]. Other measures of representational similarity are not quite this explicit but still emphasize desired *geometric invariance* properties. For example, work by Kornblith et al. [21] popularized centered kernel alignment (CKA) by emphasizing its invariance to isotropic scaling, translation, and orthogonal transformations. These are precisely the

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same invariances specified by classical shape theory [18, 10]. Earlier work by Raghu et al. [28] argued for a more flexible invariance to affine transformations, and they used canonical correlations analysis (CCA) for this purpose. Contemporary work in neuroscience is replete with similar geometric reasoning and quantitative frameworks [24].

Here we ask: What does the similarity between neural representations, potentially in terms of shape or geometry, imply about the similarity between *functions* performed by those neural systems? Potentially, not very much. Maheswaranathan et al. [26] showed that recurrent neural networks with different architectures performed the same task with similar dynamical algorithms, but with different representational geometry. More recently, Lampinen et al. [25] showed that representational geometry can be biased by other nuisance variables, such as the order in which multiple tasks are learned. Similar limitations of representational geometry measures are discussed in [11, 8].

On the other hand, several basic observations suggest that geometry and function ought to be related. Consider, for example, the common practice of using linear models to decode task-relevant variables from neural population activity [1, 23]. The premise behind these analyses is that anything linearly decodable from system A is readily accessible to layers or brain regions immediately downstream of A. Therefore, information that is linearly decodable from A may relate to functional role played by A within the overall system [4]. Notably, the invariances of typical representational similarity measures—translations, isotropic scalings, and orthogonal transformations—closely coincide with the set of transformations that leave decoding accuracy unchanged. For example, translations and rotations of neural population activity will not impact the accuracy of a linear decoder with an intercept term and an L_2 penalty on the weights. Thus, if we accept the premise that decoding accuracy is a proxy—even, perhaps, an imperfect proxy—for neural system function, then representational geometry may be a reasonable framework to capture something about this function [22].

We provide a unifying theoretical framework that connects several existing methods of measuring representational similarity in terms of linear decoding statistics. Specifically, we show that popular methods such as centered kernel alignment (CKA) [21] and similarity based on canonical correlations analysis (CCA) [28] can be interpreted as alignment scores between optimal linear decoders with particular weight regularizations. Additionally, we study how the *shape* of neural representations is related to decoding by deriving bounds that relate the average decoding distance and the Procrustes distance. We find that the Procrustes distance is a more strict notion of geometric dissimilarity than the expected distance between optimal decoder readouts, in that a small value of the Procrustes distance implies a small expected distance between optimal decoder readouts but the converse is not necessarily true. This formalizes recent observations by Cloos et al. [5], who found that high Procrustes similarity implied high CKA similarity in practical settings.

2 Theoretical Framework for Linear Decoding

We use $\boldsymbol{X} \in \mathbb{R}^{M \times N_X}$ and $\boldsymbol{Y} \in \mathbb{R}^{M \times N_Y}$ to denote matrices holding responses to M stimulus conditions measured across neural populations consisting of N_X and N_Y neurons, respectively. Throughout this paper we will assume that the neural responses have been preprocessed so that the columns of $\boldsymbol{X}$ and $\boldsymbol{Y}$ have zero mean.

2.1 Linear decoding from neural population responses

We first consider the problem of predicting (or "decoding") a target vector $\boldsymbol{z} \in \mathbb{R}^M$ from neural population responses by a linear function $\boldsymbol{X} \mapsto \boldsymbol{X}\boldsymbol{w}$ where $\boldsymbol{w} \in \mathbb{R}^{N_X}$. One can view this as a simplified neural circuit model where each element of $\hat{\boldsymbol{z}} = \boldsymbol{X}\boldsymbol{w} + \boldsymbol{\epsilon}$ is the firing rate readout unit the M conditions. Here, we interpret $\boldsymbol{w}$ as a vector of synaptic weights and the random vector $\boldsymbol{\epsilon}$ represents potential noise corruptions; for example, $\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \sigma^2)$ where $\sigma^2 > 0$ specifies the scale of noise. This neural circuit interpretation of linear decoding is fairly standard within the literature [13, 17].

Within this setting, we formalize the problem of decoding $\boldsymbol{z}$ from the population activity $\boldsymbol{X}$ through the following class of optimization problems:

$$\underset{\boldsymbol{w}}{\text{maximize}} \quad \tfrac{1}{M}\boldsymbol{z}^\top \boldsymbol{X}\boldsymbol{w} - \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{G}(\boldsymbol{X})\boldsymbol{w} \tag{1}$$

where $\boldsymbol{G}(\cdot)$ is a function mapping $\mathbb{R}^{M\times N}$ onto symmetric positive definite $N \times N$ matrices. The objective is concave, and equating the gradient to zero yields a closed form solution:

$$\boldsymbol{w}^* = \tfrac{1}{M}\boldsymbol{G}(\boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{z} \tag{2}$$

To appreciate the relevance of eq. (1), note that if $\boldsymbol{G}(\boldsymbol{X}) = \boldsymbol{C}_X + \lambda \boldsymbol{I}$, where $\boldsymbol{C}_X := \frac{1}{M}\boldsymbol{X}^\top \boldsymbol{X}$ is the empirical covariance of $\boldsymbol{X}$, then the solution $\boldsymbol{w}^*$ coincides with the optimum of the familiar ridge regression problem:

$$\underset{\boldsymbol{w}}{\text{minimize}} \quad \tfrac{1}{M}\|\boldsymbol{z} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{w}\|_2^2 \tag{3}$$

with solution $\boldsymbol{w}^* = \frac{1}{M}(\boldsymbol{C}_X + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top\boldsymbol{z}$, where $\lambda > 0$ specifies the strength of regularization on the coefficients. More generally, we can interpret eq. (1) as maximizing the inner product between the target, $\boldsymbol{z}$, and the linear readout, $\boldsymbol{X}\boldsymbol{w}$, plus a penalty term on the scale of $\boldsymbol{w}$ as quantified by the function $\boldsymbol{w} \mapsto \sqrt{\boldsymbol{w}^\top \boldsymbol{G}(\boldsymbol{X})\boldsymbol{w}}$, which is a norm on $\mathbb{R}^N$.

Equation (1) is a useful generalization of eq. (3) because there are situations where the scale of the decoded signal matters. For instance, when the readout unit is corrupted with noise, $\hat{\boldsymbol{z}} = \boldsymbol{X}\boldsymbol{w} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \sigma^2)$, we can interpret eq. (1) as maximizing the signal-to-noise ratio subject to a cost on the magnitude of the weights $\boldsymbol{w}$.

In most cases, we will consider choices of $\boldsymbol{G}(\cdot)$ that can be written as:

$$\boldsymbol{G}(\boldsymbol{X}) = a\boldsymbol{C}_X + b\boldsymbol{I} \tag{4}$$

for some $a, b \geq 0$. Under this choice, the penalty on $\boldsymbol{w}$ equals the sum of two terms:

$$\boldsymbol{w}^\top \boldsymbol{G}(\boldsymbol{X})\boldsymbol{w} = \frac{a}{M}\|\boldsymbol{X}\boldsymbol{w}\|_2^2 + b\|\boldsymbol{w}\|_2^2. \tag{5}$$

The first term, scaled by a, penalizes the activation level of the readout unit, which can be viewed as an energetic constraint. The second term, scaled by b, penalizes the strength of the synaptic weights, which can be viewed as a resource constraint for building and maintaining strong synapses. We note again that ridge regression is achieved as a special case when $a = 1$ and $b = \lambda$.

2.2 Quantifying differences between networks through the lens of decoding

We now turn our attention to the the problem of quantifying representational similarity between $\boldsymbol{X} \in \mathbb{R}^{M\times N_X}$ and $\boldsymbol{Y} \in \mathbb{R}^{M\times N_Y}$. As before, we specify a target vector $\boldsymbol{z} \in \mathbb{R}^M$ and optimize $\boldsymbol{w}$ as specified in eq. (1). We optimize $\boldsymbol{v}$ accordingly:

$$\underset{\boldsymbol{v}}{\text{maximize}} \quad \tfrac{1}{M}\boldsymbol{z}^\top\boldsymbol{Y}\boldsymbol{v} - \tfrac{1}{2}\boldsymbol{v}^\top\boldsymbol{G}(\boldsymbol{Y})\boldsymbol{v} \tag{6}$$

yielding the solution $\boldsymbol{v}^* = \frac{1}{M}\boldsymbol{G}(\boldsymbol{Y})^{-1}\boldsymbol{Y}^\top\boldsymbol{z}$. We are interested in quantifying similarity between $\boldsymbol{X}$ and $\boldsymbol{Y}$ by comparing the behavior of these optimal decoders. A straightforward way to quantify the alignment between the decoded signals is to compute their inner product, which we call the *decoding similarity* (fig. 1):

$$\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^*\rangle = \boldsymbol{w}^{*\top}\boldsymbol{X}^\top\boldsymbol{Y}\boldsymbol{v}^* = \boldsymbol{z}^\top\boldsymbol{K}_X\boldsymbol{K}_Y\boldsymbol{z} \tag{7}$$

where we have leveraged eq. (2) and the analogous closed form expression for $\boldsymbol{v}^*$ while introducing the following normalized kernel similarity matrices:

$$\boldsymbol{K}_X := \frac{1}{M}\boldsymbol{X}\boldsymbol{G}(\boldsymbol{X})^{-1}\boldsymbol{X}^\top \quad \text{and} \quad \boldsymbol{K}_Y := \frac{1}{M}\boldsymbol{Y}\boldsymbol{G}(\boldsymbol{Y})^{-1}\boldsymbol{Y}^\top. \tag{8}$$

Note that we have suppressed the dependence of $\boldsymbol{K}_X$ and $\boldsymbol{K}_Y$ on the function $\boldsymbol{G}(\cdot)$ to reduce notational clutter.

Equation (7) is a reasonable quantification of similarity between $\boldsymbol{X}$ and $\boldsymbol{Y}$ with respect to a fixed decoding task specified by $\boldsymbol{z} \in \mathbb{R}^M$, and it may be a fruitful approach for hypothesis-driven comparisons of neural systems. We comment further on this possibility in our discussion. On the other hand, many neural representations support a variety of downstream behavioral tasks. For example, features extracted in early stages of visual processing (edges and textures) can be used to support many visual tasks such as object classification, segmentation, image compression, et cetera. How can eq. (7) be adapted to quantify the similarity between $\boldsymbol{X}$ and $\boldsymbol{Y}$ across more than one pre-specified decoding task? We explore three simple ideas.